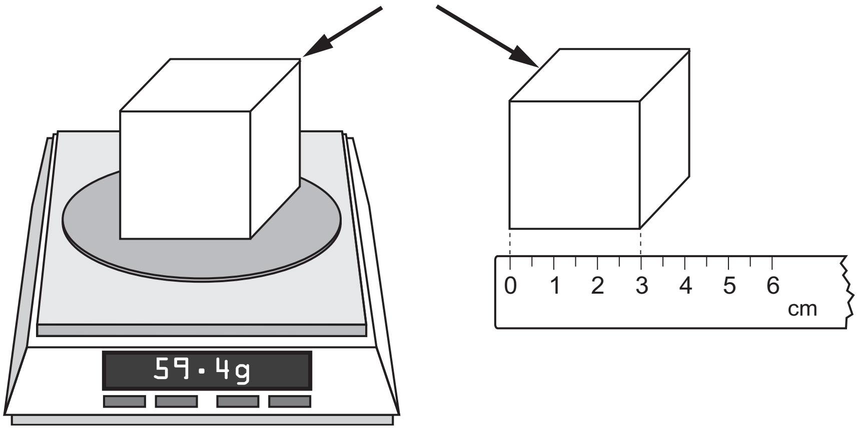
Examiner only

4. A North Wales cheese producer checks the quality of their cheese by measuring its density. This can be carried out experimentally. The two diagrams show measurements that were carried out on a cube of cheese. The balance read 0.0 g before the cube of cheese was placed on it.



(a) (i) Write down the mass of the cheese cube. .... g [1]

(ii) Write down the length of one side of the cheese cube. .... cm [1]

(iii) Use the equation:

volume of a cuboid = length × width × height

to calculate the volume of the cheese cube in cm^3. [1]

Volume = ... cm^3

(iv) Use the equation:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

to calculate the density of the cheese cube and state the unit. [3]

Density = ... Unit = ...

Examiner only

(b) (i) A worker at the cheese factory suggests that the mass of the cheese cube should be measured more than once. State **two** reasons why this is good scientific practice. [2]

....

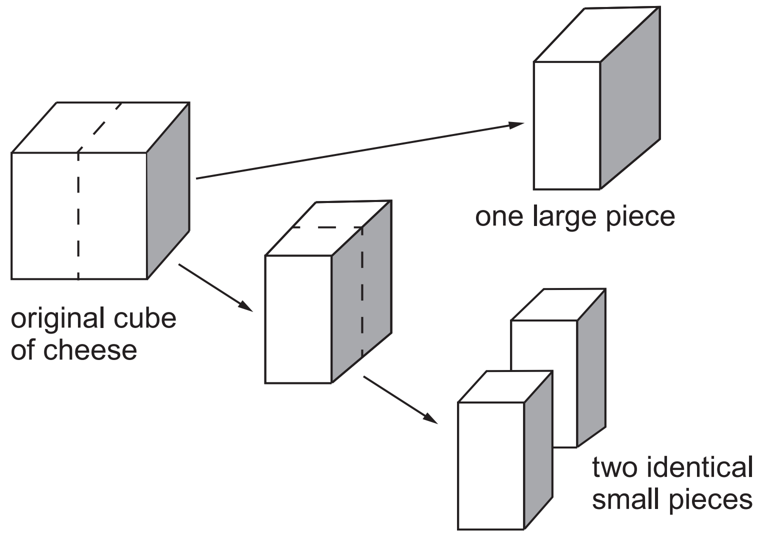
....

(ii) State how the **measurements** could be improved. [1]

....

....

(c) The cube of cheese used in the experiment is now cut into two identical halves. One of these pieces is then cut in half. The sample of cheese has been divided into three pieces, one large piece and two identical small pieces.



Tick (✓) the **two** correct statements. [2]

One of the small pieces of cheese is $\frac{1}{3}$ of the mass of the large piece.	☐
One of the small pieces of cheese is $\frac{1}{3}$ of the volume of the large piece.	☐
The three pieces of cheese have the same density.	☐
The mass of the large piece of cheese is half the mass of the original cube.	☐
The mass to volume ratio for each of the three samples is different.	☐

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